Lab Exam:01

Title: Association/Classification/Fact and Dimension Table

Course title: ***Data Mining Problem Description***

Course code: CSE-452
4^{th} Year 1^{st} Semester Examination 2025

Date of Submission: 23/April/2026

Submitted to-

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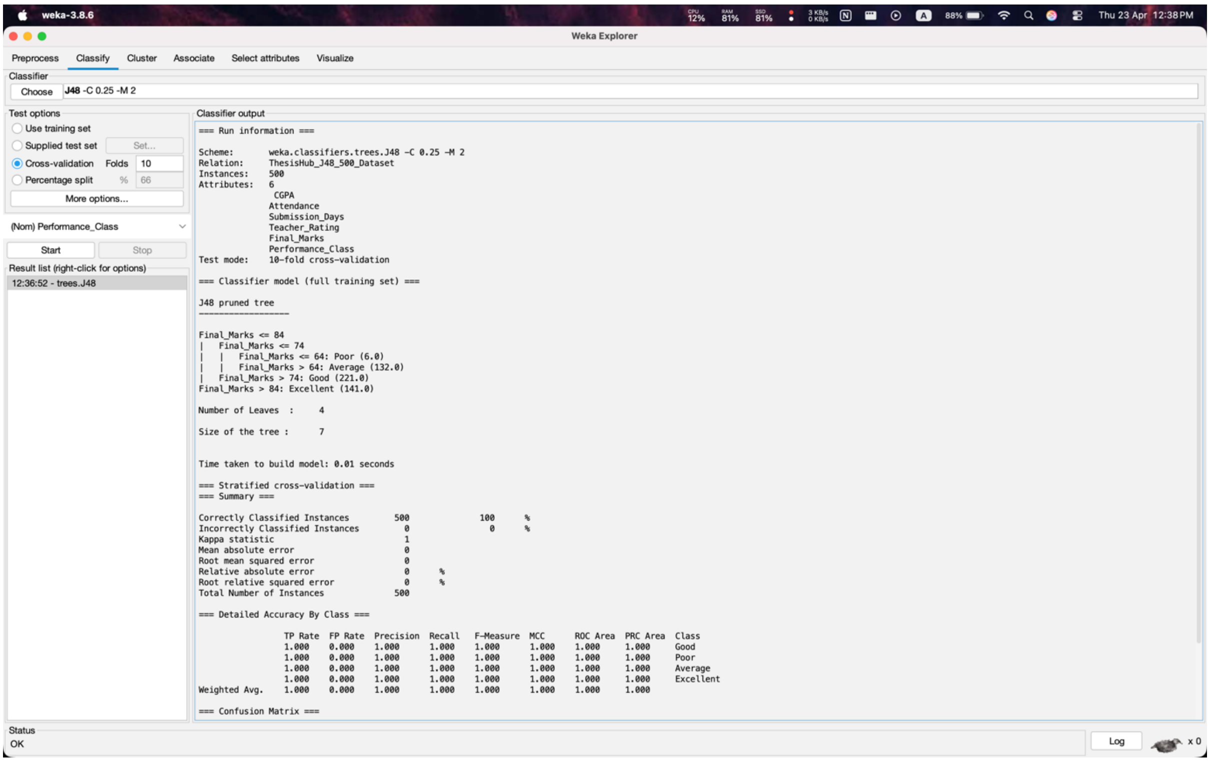
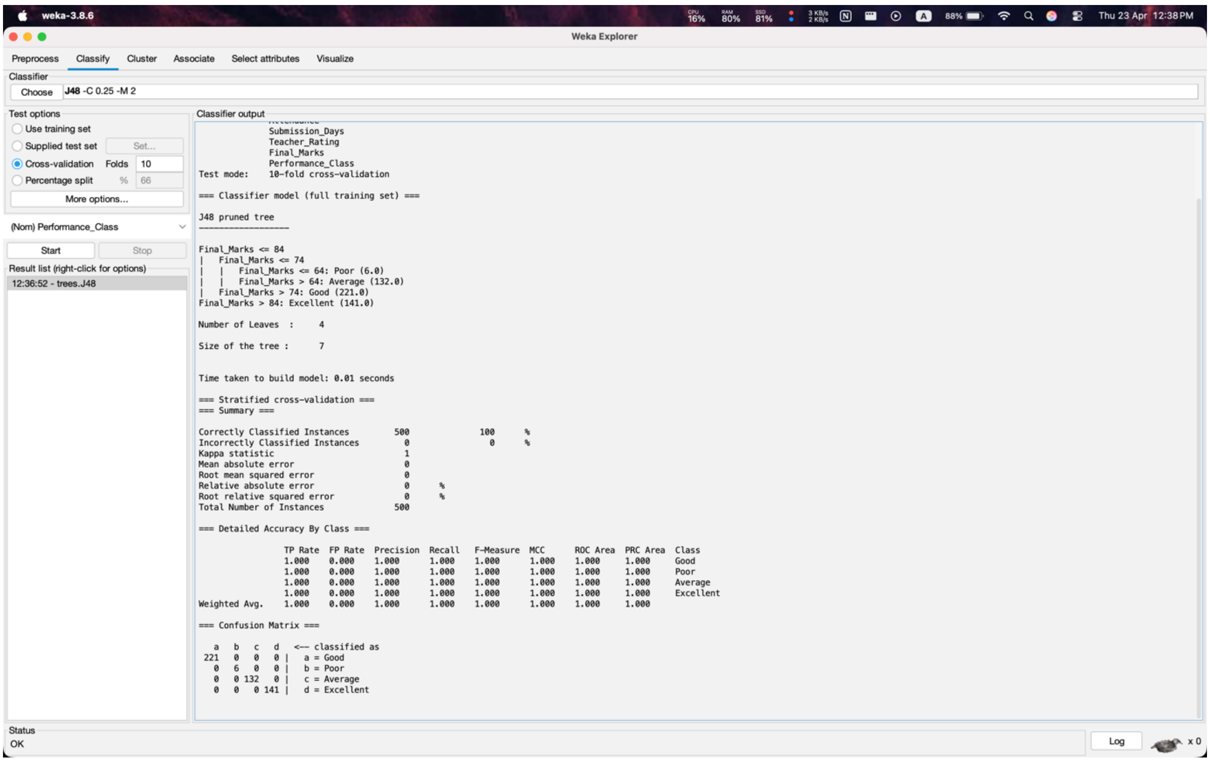
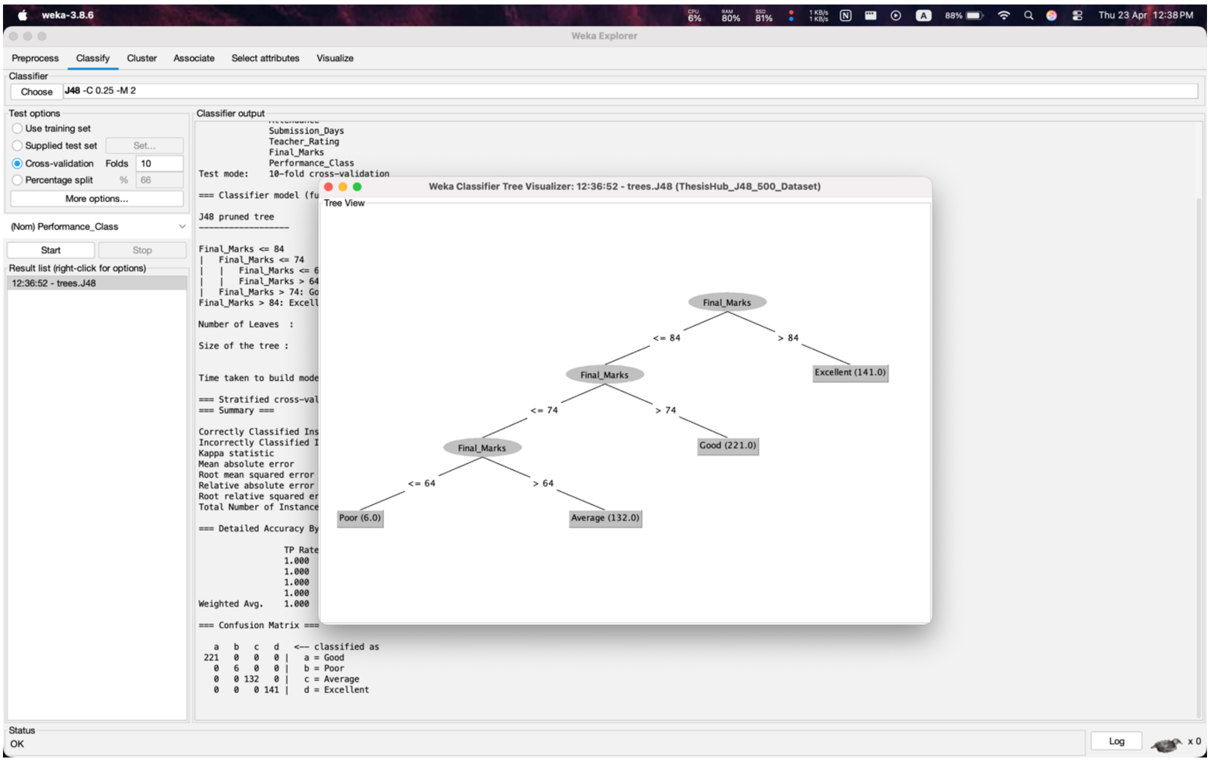
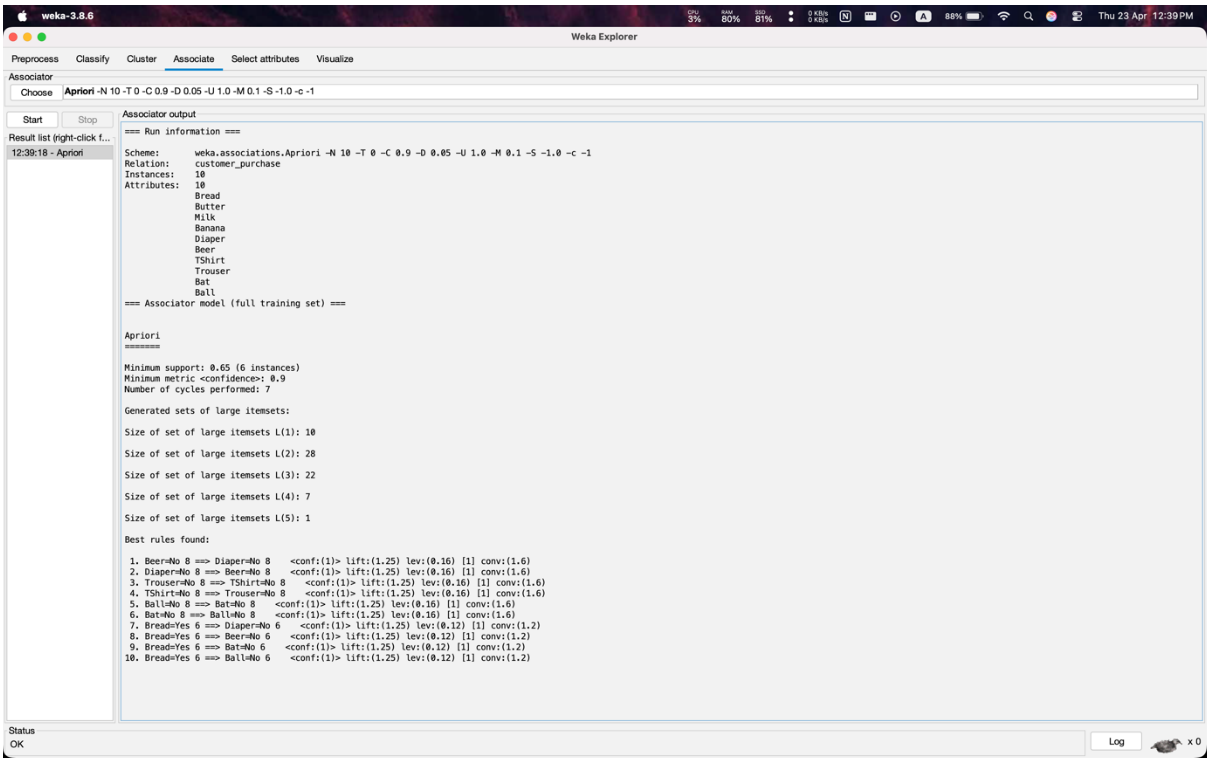
1. Classification On my project dataset “Intelligence Automatic Supervisor Allocation System ”







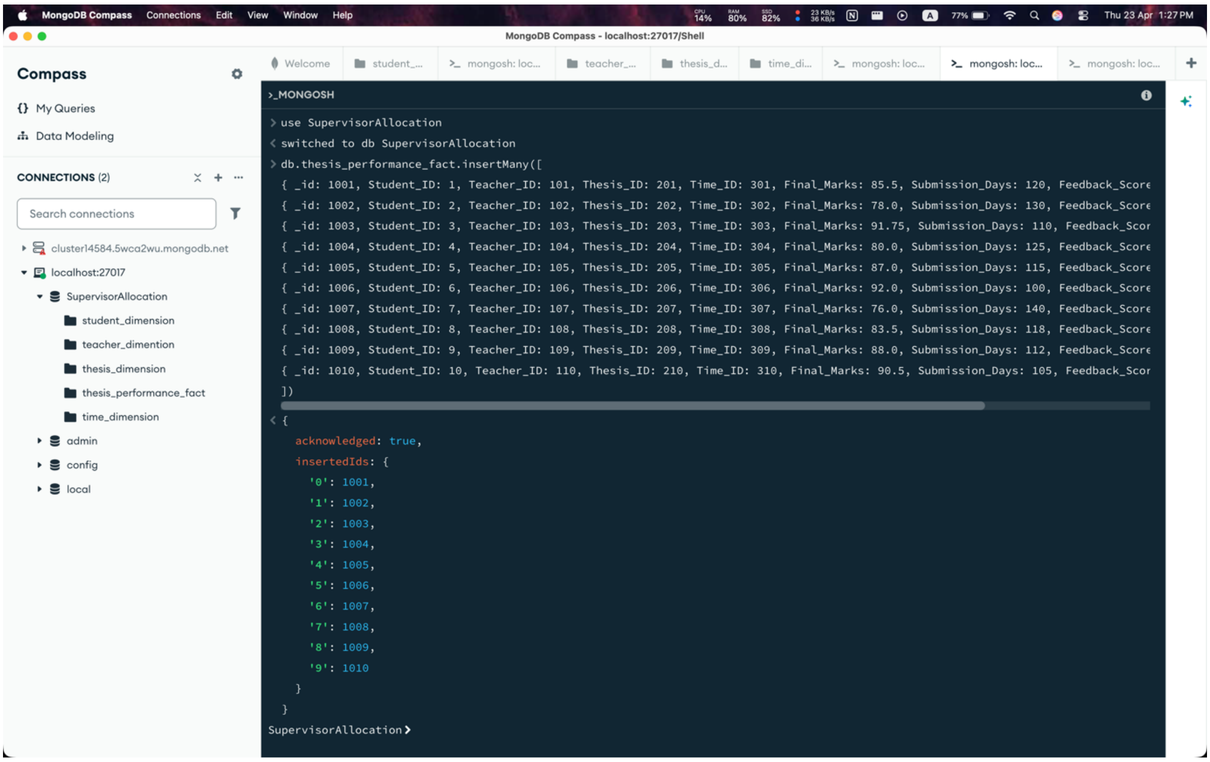
2. **Association Rule On my project dataset "Intelligence Automatic Supervisor Allocation System"**



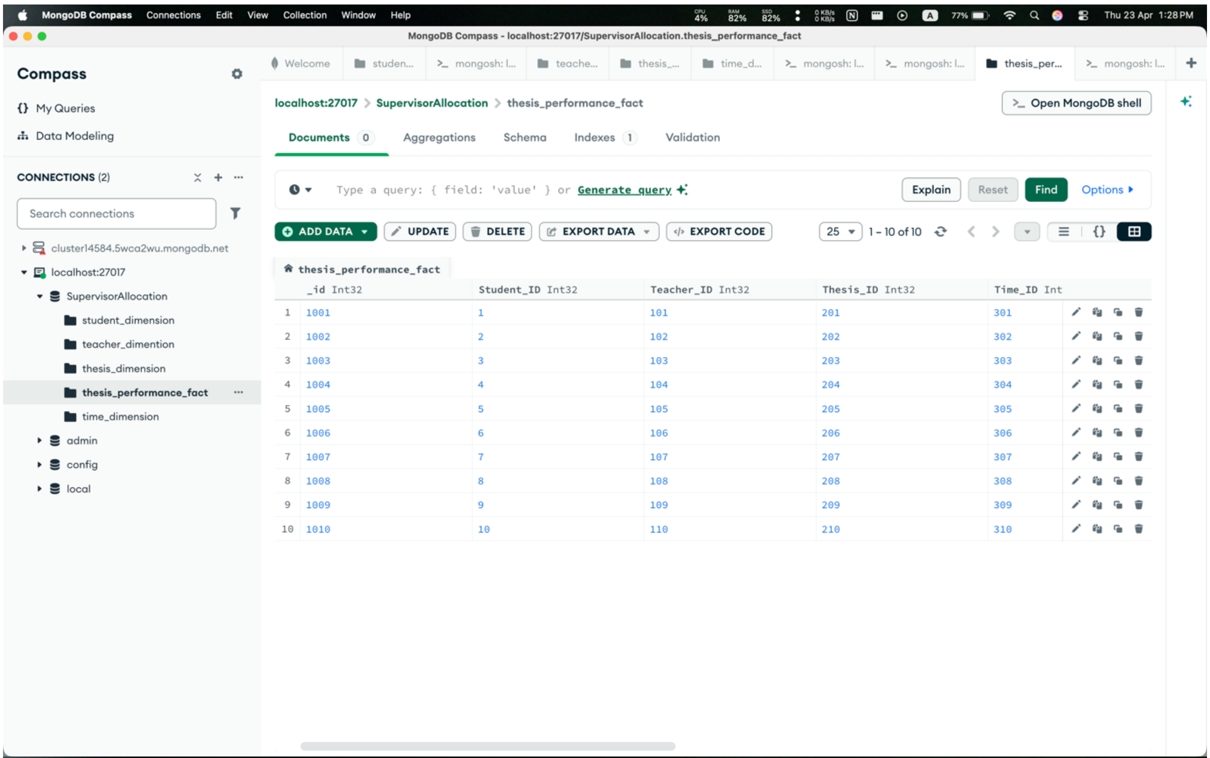
3. Design Fact Table and Dimension Table

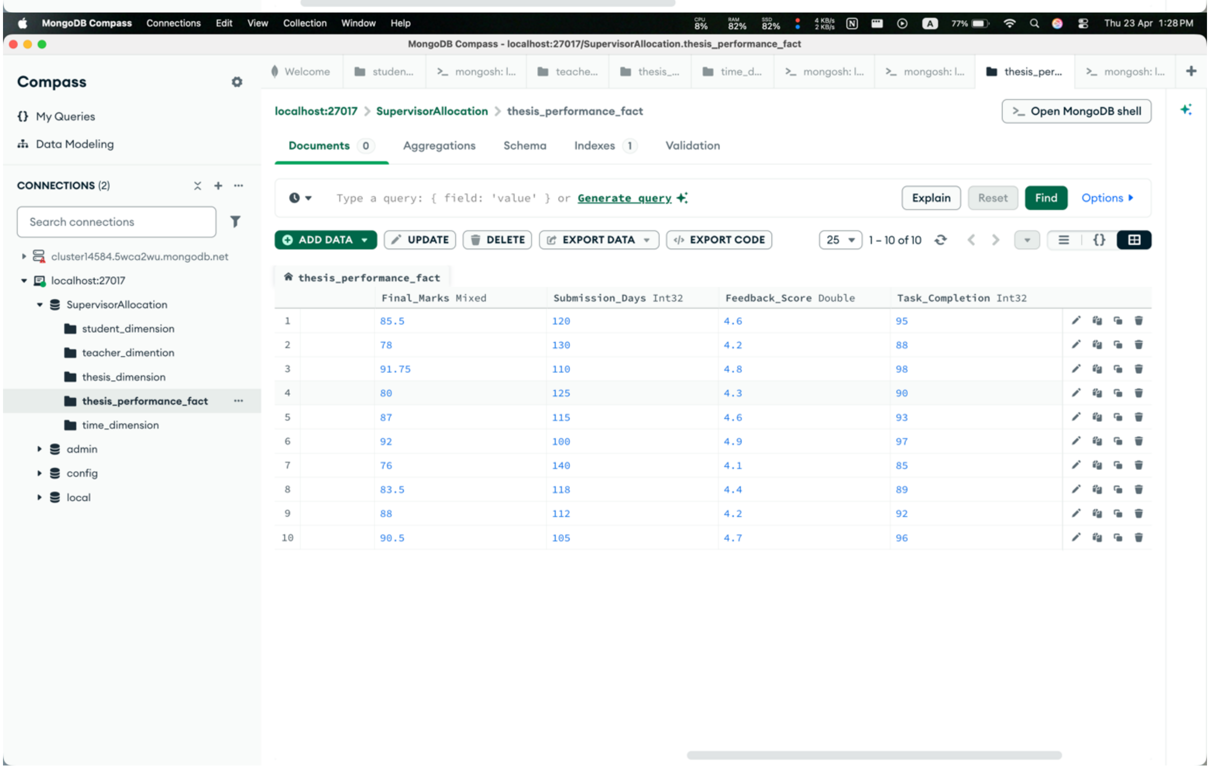
Fact Table:

1. Creation



2. Shows data


MongoDB Compass - localhost:27017/SupervisorAllocation.thesis_performance_fact
Compass
My Queries
Data Modeling
CONNECTIONS (2)
Search connections
cluster14584.5wca2wu.mongodb.net
localhost:27017
SupervisorAllocation
student_dimension
teacher_dimention
thesis_dimension
thesis_performance_fact
time_dimension
admin
config
local
localhost:27017 > SupervisorAllocation > thesis_performance_fact
Open MongoDB shell
Documents 0
Aggregations
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
25
1 – 10 of 10
thesis_performance_fact
_id Int32 | Student_ID Int32 | Teacher_ID Int32 | Thesis_ID Int32 | Time_ID Int
1 1001 1 101 201 301
2 1002 2 102 202 302
3 1003 3 103 203 303
4 1004 4 104 204 304
5 1005 5 105 205 305
6 1006 6 106 206 306
7 1007 7 107 207 307
8 1008 8 108 208 308
9 1009 9 109 209 309
10 1010 10 110 210 310


MongoDB Compass - localhost:27017/SupervisorAllocation.thesis_performance_fact
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time_dimension
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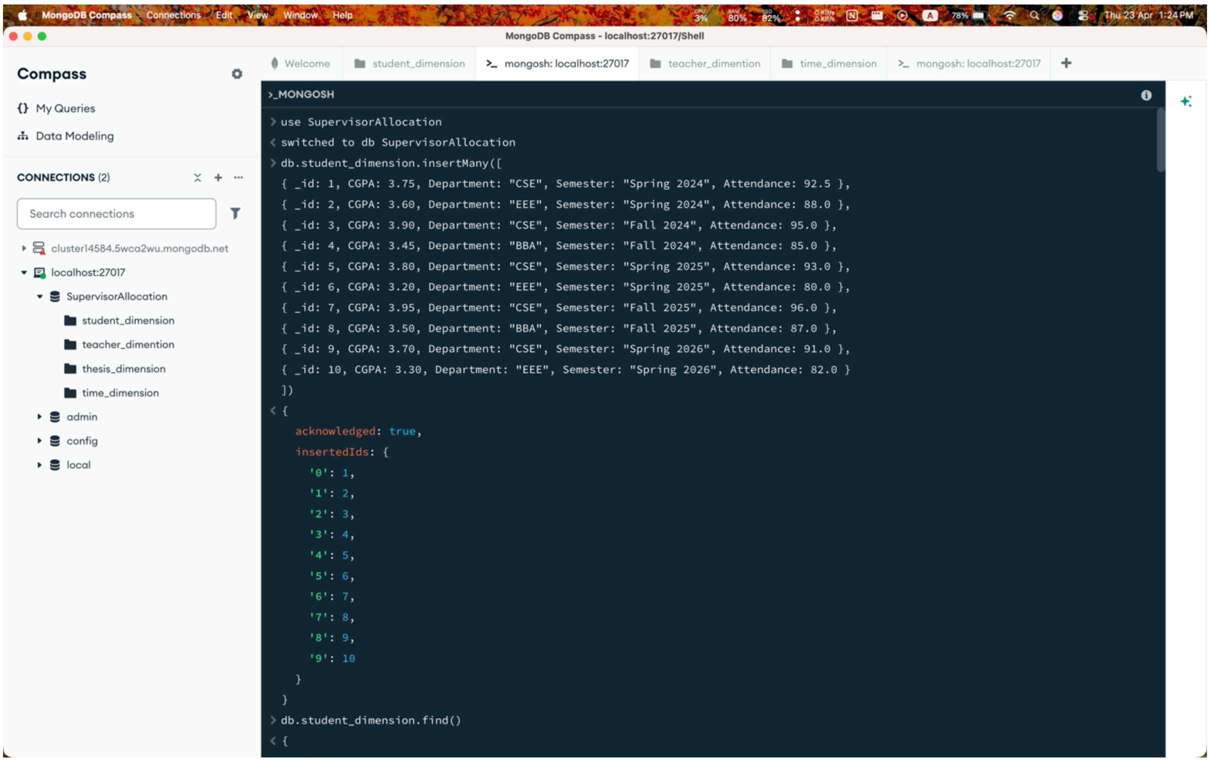
Schema
Indexes 1
Validation
Type a query: { field: 'value' } or Generate query
Explain
Reset
Find
Options
ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE
25
1 – 10 of 10
thesis_performance_fact
Final_Marks Mixed | Submission_Days Int32 | Feedback_Score Double | Task_Completion Int32
1 85.5 120 4.6 95
2 78 130 4.2 88
3 91.75 110 4.8 98
4 80 125 4.3 90
5 87 115 4.6 93
6 92 100 4.9 97
7 76 140 4.1 85
8 83.5 118 4.4 89
9 88 112 4.2 92
10 90.5 105 4.7 96

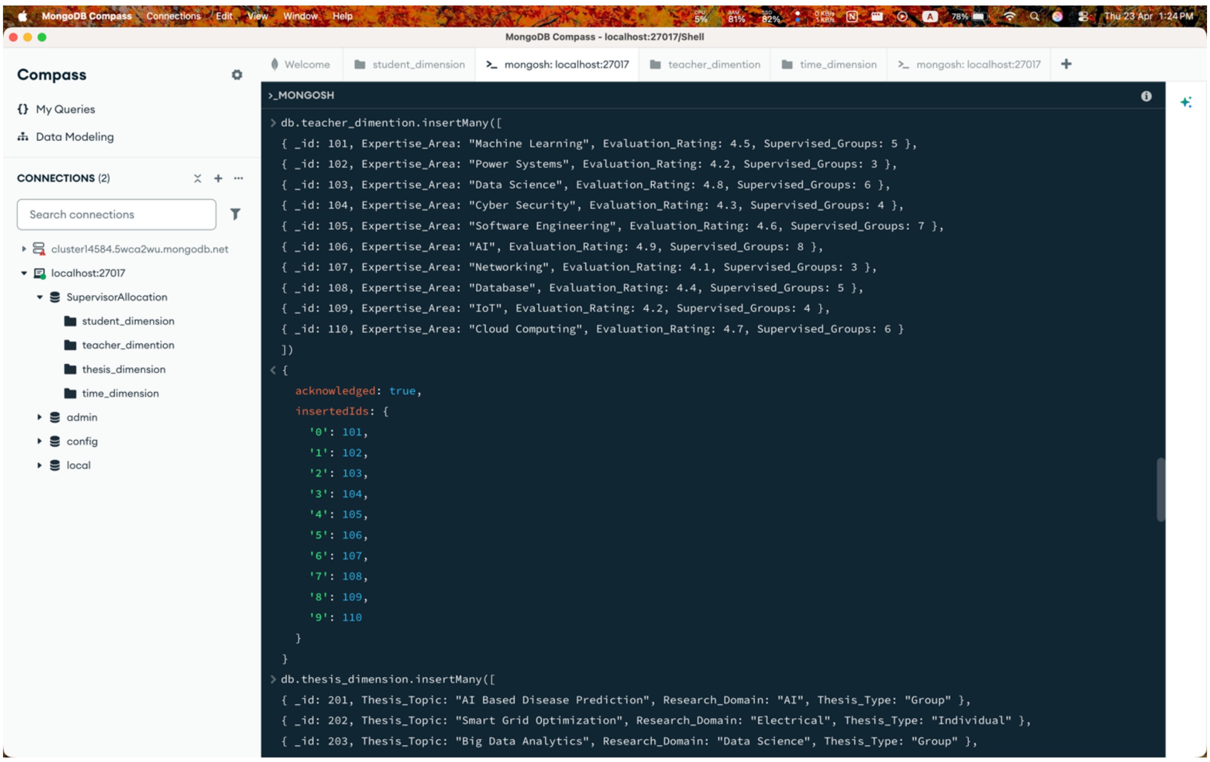
Dimension Table

1. Creation

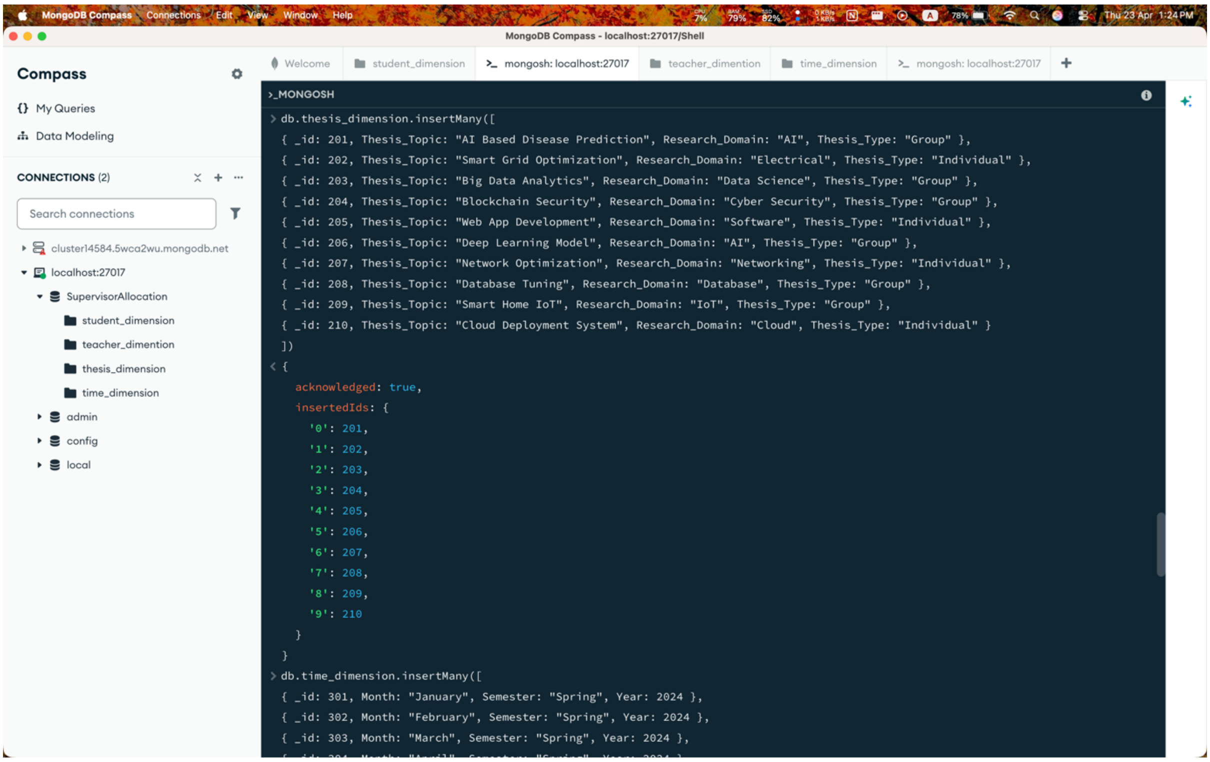
1.1 Student_dimension



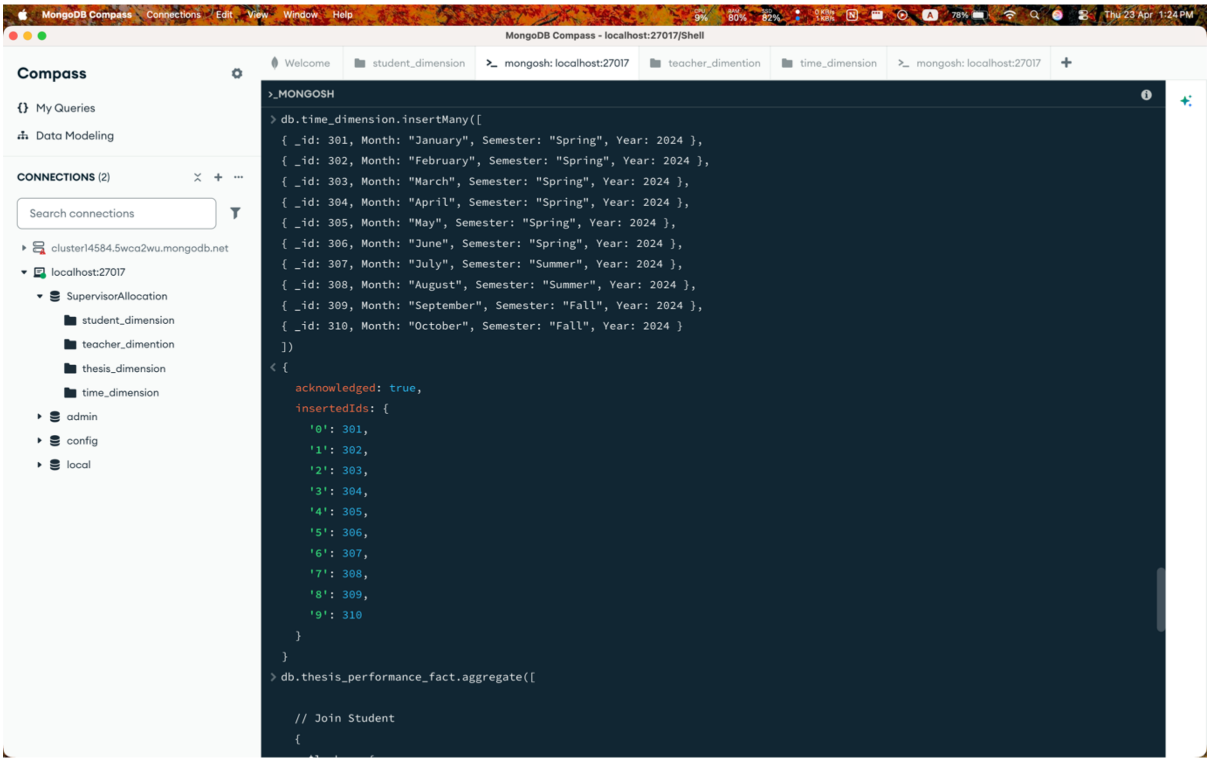
1.2 Teacher_dimension



1.3 Thesis_dimension

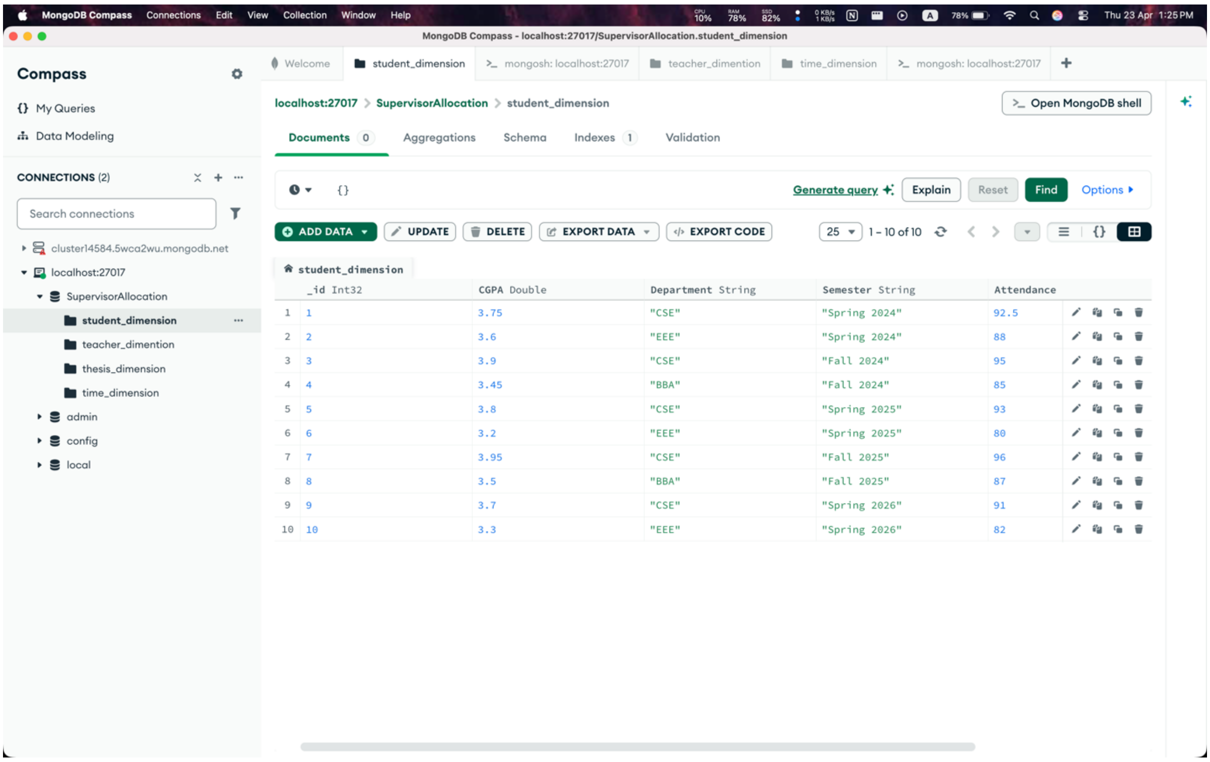


1.4 Time_dimension



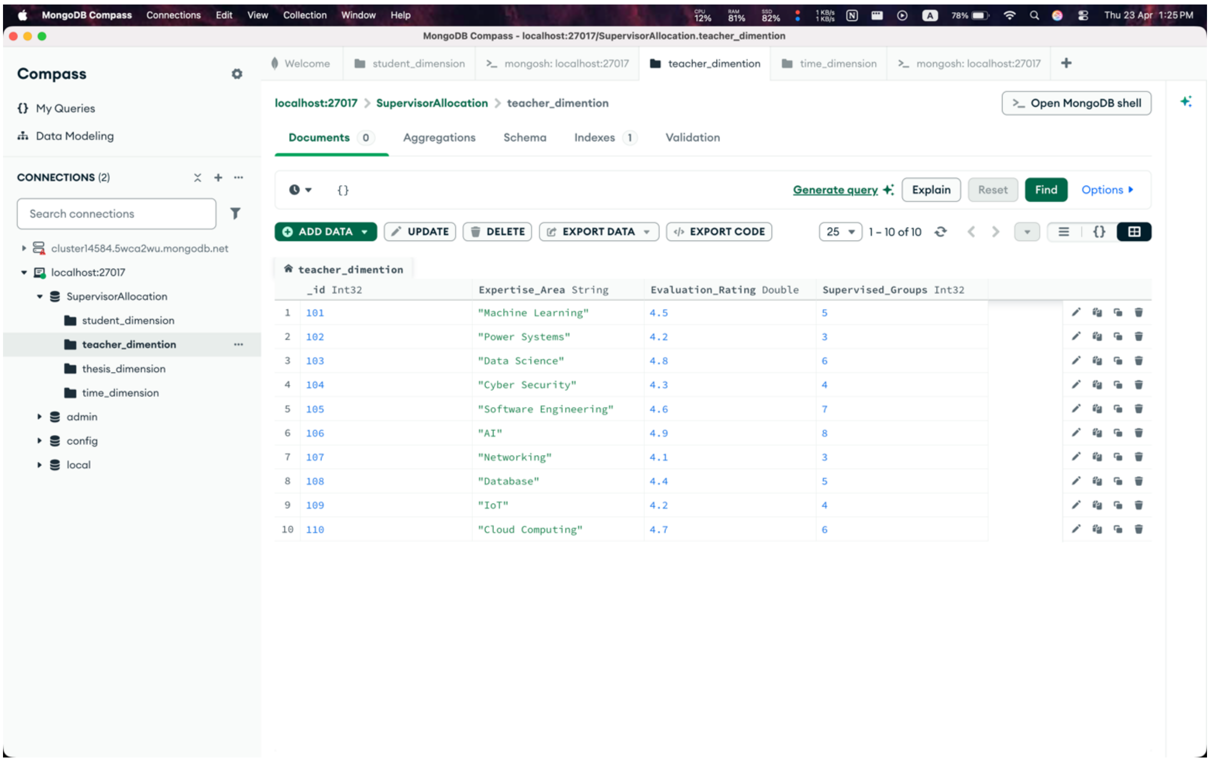
Show all dimension data:

1.1 Student_dimension Table



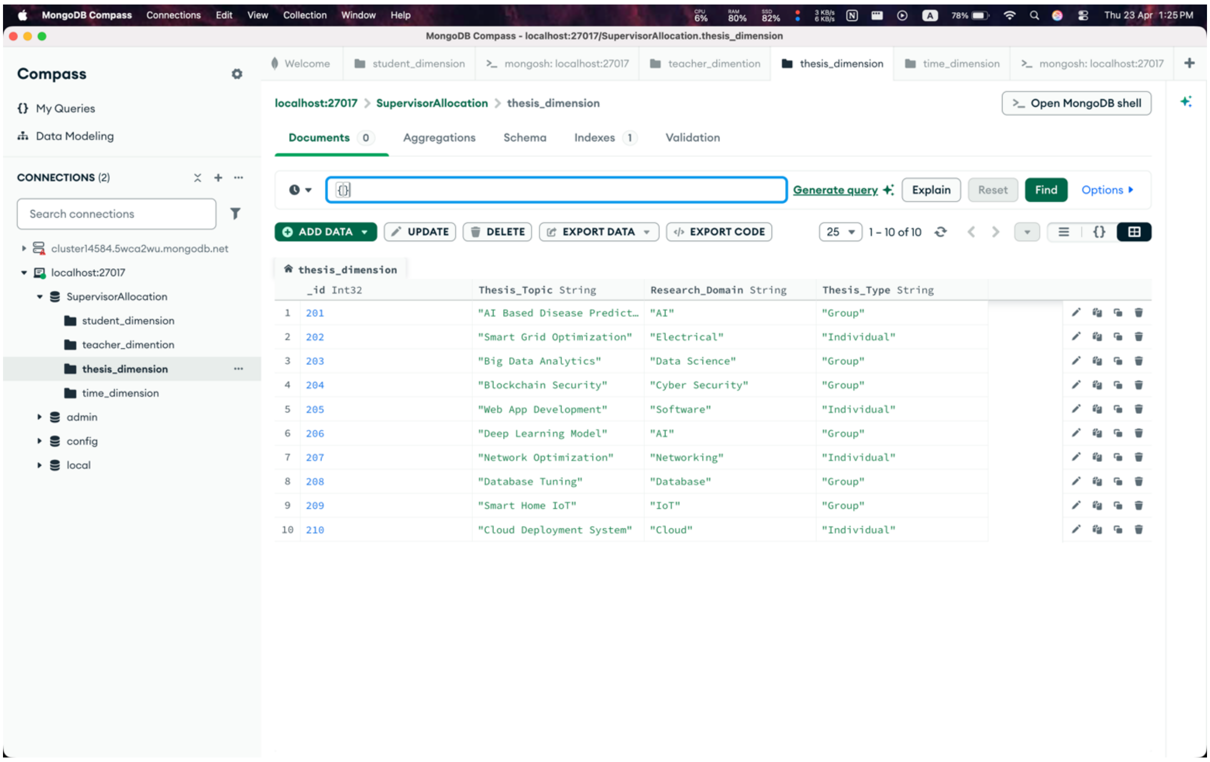
_id Int32	CGPA Double	Department String	Semester String	Attendance
1	3.75	"CSE"	"Spring 2024"	92.5
2	3.6	"EEE"	"Spring 2024"	88
3	3.9	"CSE"	"Fall 2024"	95
4	3.45	"BBA"	"Fall 2024"	85
5	3.8	"CSE"	"Spring 2025"	93
6	3.2	"EEE"	"Spring 2025"	80
7	3.95	"CSE"	"Fall 2025"	96
8	3.5	"BBA"	"Fall 2025"	87
9	3.7	"CSE"	"Spring 2026"	91
10	3.3	"EEE"	"Spring 2026"	82

1.2 Teacher_dimension Table



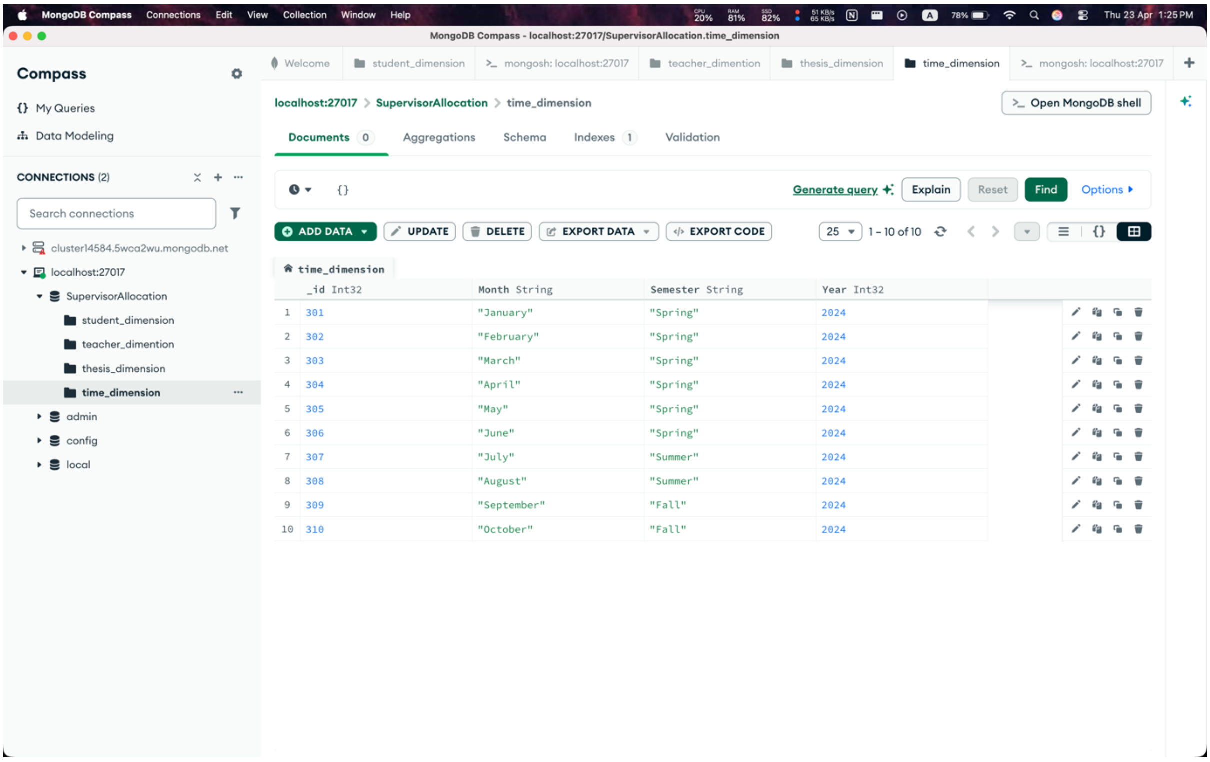
_id Int32	Expertise_Area String	Evaluation_Rating Double	Supervised_Groups Int32
101	"Machine Learning"	4.5	5
102	"Power Systems"	4.2	3
103	"Data Science"	4.8	6
104	"Cyber Security"	4.3	4
105	"Software Engineering"	4.6	7
106	"AI"	4.9	8
107	"Networking"	4.1	3
108	"Database"	4.4	5
109	"IoT"	4.2	4
110	"Cloud Computing"	4.7	6

1.3 Thesis_dimension Table



_id Int32	Thesis_Topic String	Research_Domain String	Thesis_Type String
201	"AI Based Disease Predict…	"AI"	"Group"
202	"Smart Grid Optimization"	"Electrical"	"Individual"
203	"Big Data Analytics"	"Data Science"	"Group"
204	"Blockchain Security"	"Cyber Security"	"Group"
205	"Web App Development"	"Software"	"Individual"
206	"Deep Learning Model"	"AI"	"Group"
207	"Network Optimization"	"Networking"	"Individual"
208	"Database Tuning"	"Database"	"Group"
209	"Smart Home IoT"	"IoT"	"Group"
210	"Cloud Deployment System"	"Cloud"	"Individual"

1.4 Time_dimension Table



_id Int32	Month String	Semester String	Year Int32
301	"January"	"Spring"	2024
302	"February"	"Spring"	2024
303	"March"	"Spring"	2024
304	"April"	"Spring"	2024
305	"May"	"Spring"	2024
306	"June"	"Spring"	2024
307	"July"	"Summer"	2024
308	"August"	"Summer"	2024
309	"September"	"Fall"	2024
310	"October"	"Fall"	2024